RAG Learning

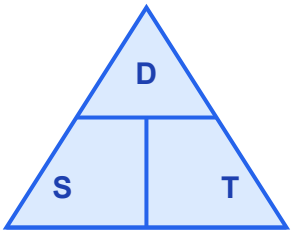
3.2 Rates of Change and Compound Measures

OCR GCSE Mathematics J560 | Component 3 (Ratio, Proportion and Rates of Change) | Papers 1, 2 & 3

3.2.1 Speed, Distance and Time

Speed is a compound measure — it combines distance and time. **Average speed** is the total distance divided by the total time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} \qquad \text{Distance} = \text{Speed} \times \text{Time} \qquad \text{Time} = \frac{\text{Distance}}{\text{Speed}}$$



SDT Triangle
Cover the required quantity

Common Units

Speed	Distance	Time
m/s (metres per second)	metres (m)	seconds (s)
km/h (kilometres per hour)	kilometres (km)	hours (h)

mph (miles per hour)	miles	hours (h)

Exam Tip: Convert time to a decimal (not minutes) before calculating. 1 hour 30 mins = 1.5 hours, NOT 1.3 hours.

Worked Example: A car travels 135 km in 1 hour 30 minutes. Find the average speed in km/h.

Time = 1.5 h
Speed = $135 \div 1.5 = \mathbf{90}\,\mathrm{km/h}$

Worked Example: A train travels at 80 km/h for 45 minutes. How far does it travel?

Time = $45 \div 60 = 0.75\,\mathrm{h}$
Distance = $80 \times 0.75 = \mathbf{60}\,\mathrm{km}$

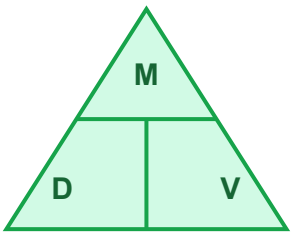
Worked Example: Convert 20 m/s to km/h.

$20\,\mathrm{m/s} \times 3600\,\mathrm{s/h} = 72{,}000\,\mathrm{m/h} = \mathbf{72}\,\mathrm{km/h}$

3.2.2 Density, Mass and Volume

Density is the mass per unit volume of a substance. A denser material has more mass packed into the same volume.

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \qquad \text{Mass} = \text{Density} \times \text{Volume} \qquad \text{Volume} = \frac{\text{Mass}}{\text{Density}}$$



DMV Triangle
Cover the required quantity

Common Units

Density	Mass	Volume
g/cm³ (grams per cm³)	grams (g)	cm³
kg/m³	kilograms (kg)	m³

Worked Example: A block of iron has mass 790 g and volume 100 cm³. Find its density.

Density = $790 \div 100 = \mathbf{7.9}\,\mathrm{g/cm^3}$

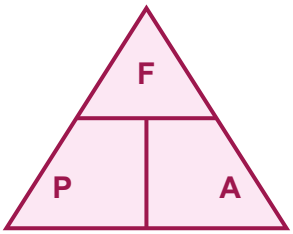
Worked Example: Gold has density 19.3 g/cm³. Find the volume of a 96.5 g gold bar.

Volume = Mass ÷ Density = $96.5 \div 19.3 = \mathbf{5}\,\mathrm{cm^3}$

3.2.3 Pressure, Force and Area

Pressure is the force applied per unit area. The greater the force or the smaller the area, the higher the pressure.

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} \qquad \text{Force} = \text{Pressure} \times \text{Area} \qquad \text{Area} = \frac{\text{Force}}{\text{Pressure}}$$



Pressure–Force–Area Triangle

Units

- Pressure: **Pascals (Pa)** = N/m^2 (Newtons per square metre)
- Force: **Newtons (N)**
- Area: **m^2** (or cm^2)

Worked Example: A force of 600 N acts on an area of 0.3 m^2. Find the pressure.

Pressure = $600 \div 0.3 = \mathbf{2000}\,\text{Pa}$

Worked Example: A box exerts a pressure of 250 Pa on the ground. The area of the base is 0.8 m^2. Find the force.

Force = $250 \times 0.8 = \mathbf{200}\,\text{N}$

Exam Tip: Check units are consistent before substituting. If area is given in cm^2, convert to m^2 (divide by 10,000) if pressure is in Pa.

3.2.4 Percentage Change and Growth/Decay

A **multiplier** is the number you multiply by to apply a percentage change in one step.

- Increase by r%: multiply by $\left(1 + \frac{r}{100}\right)$
- Decrease by r%: multiply by $\left(1 - \frac{r}{100}\right)$

Change	Multiplier	Example
Increase by 20%	1.20	£50 → £60
Decrease by 15%	0.85	£200 → £170
Increase by 7.5%	1.075	
Decrease by 3%	0.97	

Repeated Percentage Change

Apply the multiplier n times by raising it to the power n.

Worked Example: A car's value is £12,000. It depreciates by 8% each year. Find its value after 3 years.

Multiplier = $1 - 0.08 = 0.92$
Value = $12{,}000 \times 0.92^3 = 12{,}000 \times 0.778688 = \boldsymbol{£9{,}344.26}$ (2 d.p.)

Worked Example: A population of 5000 increases by 4% per year. What will the population be after 5 years?

Multiplier = 1.04
Population = $5000 \times 1.04^5 = 5000 \times 1.21665\ldots \approx \mathbf{6083}$

3.2.5 Compound Interest and Depreciation

Compound Interest:

$$A = P\left(1 + \frac{r}{100}\right)^n$$

Depreciation:

$$A = P\left(1 - \frac{r}{100}\right)^n$$

where P = principal (starting amount), r = percentage rate per period, n = number of periods, A = final amount.

Worked Example: £5,000 is invested at 3.5% compound interest per year. Find the total amount after 4 years.

$$A = 5000 \times \left(1 + \frac{3.5}{100}\right)^4 = 5000 \times 1.035^4$$

$1.035^4 = 1.14752\ldots$
$A = 5000 \times 1.14752 = \boldsymbol{£5{,}737.60}$ (2 d.p.)

Worked Example: A machine costs £24,000 and depreciates at 12% per year. Find its value after 6 years.

$$A = 24{,}000 \times (1 - 0.12)^6 = 24{,}000 \times 0.88^6$$

$0.88^6 = 0.46440\ldots$
$A = 24{,}000 \times 0.46440 = \boldsymbol{£11{,}145.60}$ (2 d.p.)

Worked Example: Find the compound interest earned (not total amount) when £8,000 is invested at 2% per annum for 3 years.

$A = 8000 \times 1.02^3 = 8000 \times 1.0612 = £8{,}489.60$

Interest = $£8{,}489.60 - £8{,}000 = \mathbf{£489.60}$

Exam Tip: Compound interest means interest is calculated on the new total each period (interest on interest). Simple interest only applies the rate to the original principal each time.

Comparing Compound and Simple Interest

Type	Formula	Comment
Simple interest	$I = \dfrac{P \times r \times t}{100}$	Interest same every period
Compound interest	$A = P(1 + r/100)^n$	Interest grows each period

3.2.6 Rates of Change from Graphs

Distance–Time Graphs

The **gradient** of a distance–time graph equals the **speed** at that point.

$$\text{Speed} = \frac{\text{change in distance}}{\text{change in time}} = \frac{\Delta d}{\Delta t}$$

A **horizontal line** means the object is stationary (speed = 0). A **steeper line** means faster speed. A **negative gradient** means the object is returning towards the start.

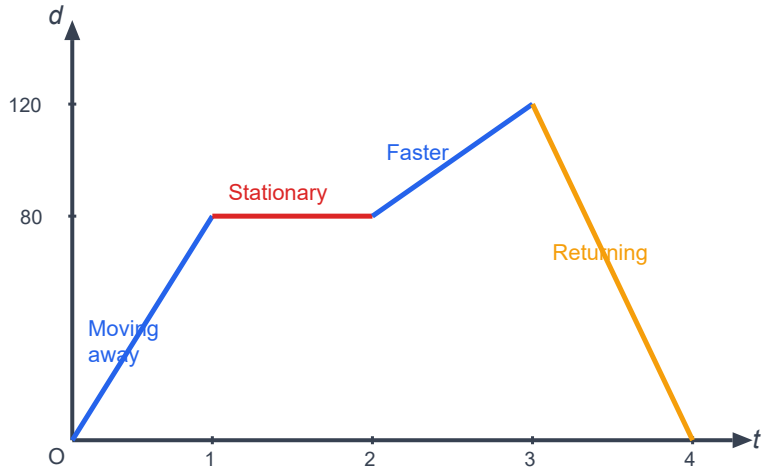


Fig 1 — Distance–time graph. Blue = moving, red = stationary, amber = returning. Gradient = speed.

Speed–Time Graphs

- The **gradient** of a speed–time graph = **acceleration** (change in speed per unit time).
- The **area under** a speed–time graph = **distance travelled**.
- A horizontal line means **constant speed** (zero acceleration).
- A negative gradient means the object is decelerating (slowing down).

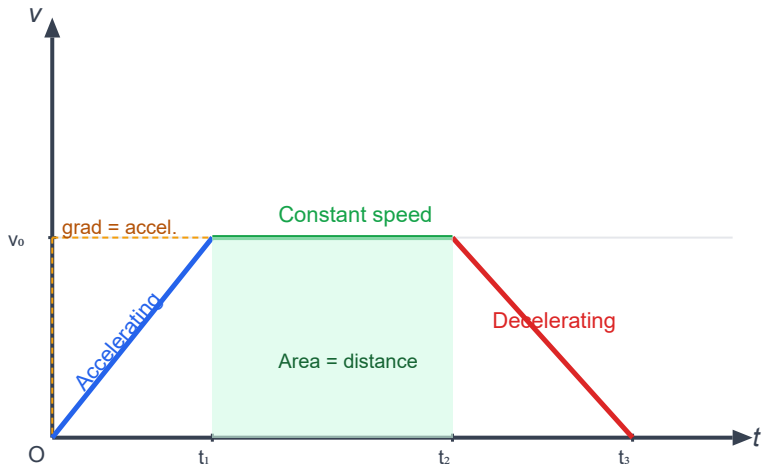


Fig 2 — Speed–time graph. Gradient = acceleration. Area under graph = distance.

Worked Example: A speed–time graph shows a vehicle accelerating from 0 to 20 m/s over 5 s, then travelling at constant 20 m/s for 10 s, then decelerating to rest in 4 s.
Find: (a) the acceleration, (b) the total distance travelled.

(a) Acceleration = gradient = $(20-0)/(5-0) = \mathbf{4}\,\mathrm{m/s^2}$

(b) Distance = area under graph
— Accelerating: $\frac{1}{2} \times 5 \times 20 = 50\,\mathrm{m}$ (triangle)
— Constant: $10 \times 20 = 200\,\mathrm{m}$ (rectangle)
— Decelerating: $\frac{1}{2} \times 4 \times 20 = 40\,\mathrm{m}$ (triangle)
Total distance = $50 + 200 + 40 = \mathbf{290}\,\mathrm{m}$

Exam Tip: For curved speed–time graphs, estimate the area by counting squares or by dividing into trapezoids.

Formula Summary

Topic	Key Formula	Units
Speed	$S = D/T$	m/s, km/h, mph
Density	$\rho = m/V$	g/cm³, kg/m³
Pressure	$P = F/A$	Pa = N/m²
% increase multiplier	$\times(1 + r/100)$	—
% decrease multiplier	$\times(1 - r/100)$	—
Compound interest	$A = P(1 + r/100)^n$	£, $, €
Depreciation	$A = P(1 - r/100)^n$	£, $, €
Gradient of d–t graph	Speed	m/s, km/h
Gradient of v–t graph	Acceleration	m/s²
Area under v–t graph	Distance	m, km

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